

Selenium Automation Testing Report

Name:Karthikeyan S

Date:29-06-2026

Title

Purplle Website Automation Testing using Selenium WebDriver

1. Introduction

The automation scenario focuses on validating the product search and add-to-cart functionality in the Purplle e-commerce website.

Website Tested:

<https://www.purplle.com>

2. Objective

As a user, the objective is to:

- Search for beauty products
- Validate product details
- Add product to cart
- Validate cart functionality

Automation validates the complete user journey from search to cart.

3. Tools & Technologies Used

Tool / Technology	Purpose
Java 22	Programming Language
Selenium WebDriver	Browser Automation
Maven	Dependency Management
JUnit 5	Test Framework
ChromeDriver	Browser Driver

4. Framework Design Pattern

The project uses:

Page Object Model (POM)

Benefits:

- Reusable code
 - Better maintainability
 - Cleaner test scripts
 - Separation of pages and tests
 - Easy scalability
-

5. Project Folder Structure

week3Project

src/test/java/com/example/Selenium/config/Config.java

src/test/java/com/example/Selenium/pages/HomePage.java

src/test/java/com/example/Selenium/pages/SearchResultsPage.java

src/test/java/com/example/Selenium/pages/ProductPage.java

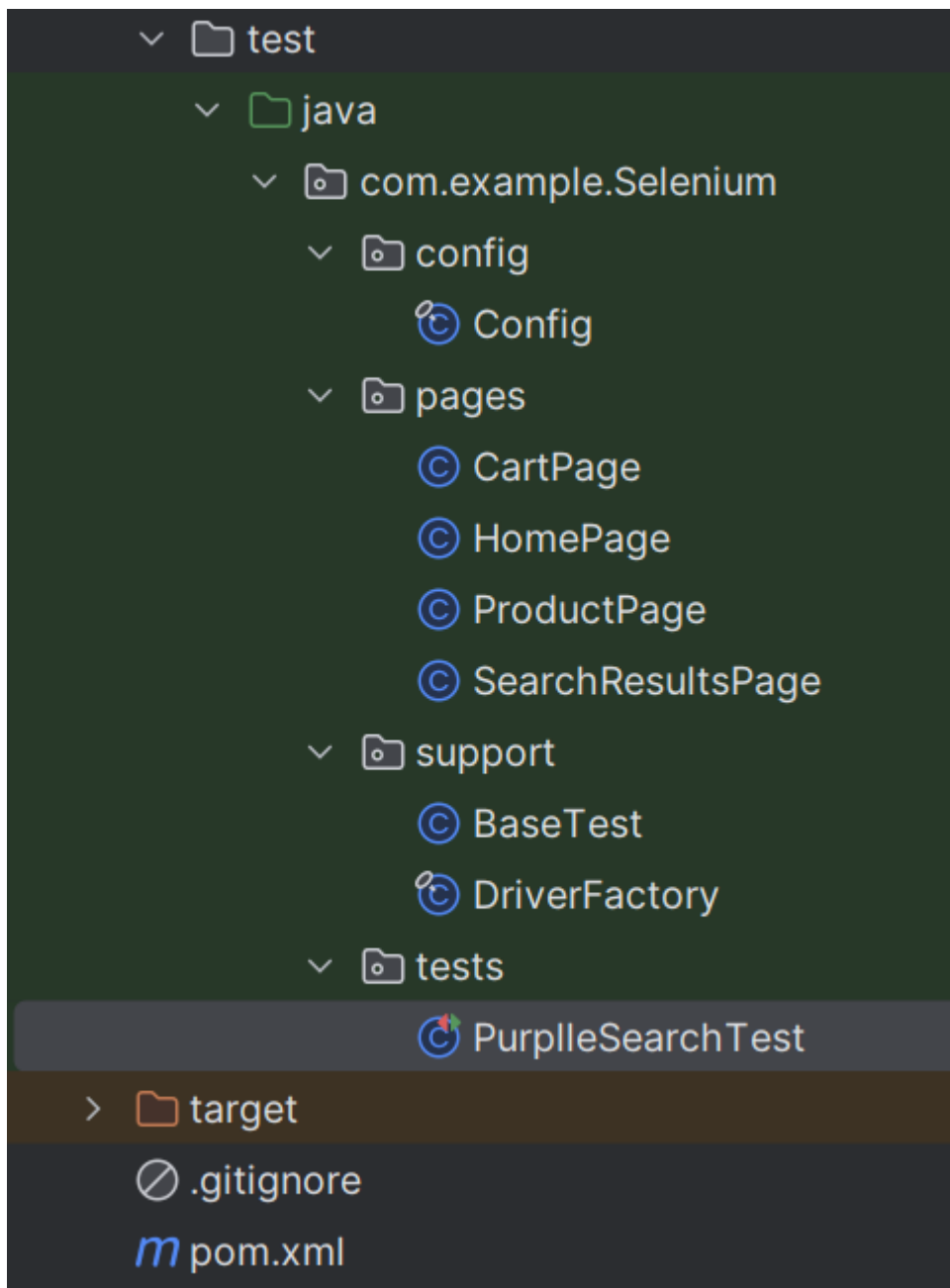
src/test/java/com/example/Selenium/pages/CartPage.java

src/test/java/com/example/Selenium/support/BaseTest.java

src/test/java/com/example/Selenium/support/DriverFactory.java

src/test/java/com/example/Selenium/tests/PurpleSearchTest.java

pom.xml



6. Test Scenario

Scenario Name

Search Product and Add Product to Cart

7. Test Flow

Step 1

Launch Chrome Browser

Step 2

Navigate to:

<https://www.purple.com>

Step 3

Locate search bar

Step 4

Search for:

lipstick

Step 5

Verify search results page loads

Step 6

Capture and print all products dynamically

Step 7

Click first available product dynamically

Step 8

Validate:

- Product Name
- Product Price
- Product Rating

Step 9

Click:

Add to Cart

Step 10

Validate cart page is displayed

Step 11

Validate cart count increment

8. Automation Expectations Covered

Requirement	Status
Dynamic Locators	Implemented
Product Iteration	Implemented
Add to Cart Validation	Implemented
Product Validation	Implemented
Rating Validation	Implemented
Search Functionality	Implemented
Cart Validation	Implemented

9. Edge Cases Covered

Edge Case	Handling
No Search Results	Dynamic wait
Product without Rating	Returns "Rating Not Available"
Multiple Similar Products	Iteration implemented
Add-to-cart Delay	Explicit waits used

10. Config.java

Purpose:

- Stores reusable configuration values
- Maintains Base URL
- Controls Headless Execution

Key Features:

- Base URL setup
- Headless mode support

```
package com.example.Selenium.config;

public final class Config { no usages

    private Config(){} no usages

    public static String baseUrl() { no usages
        return "https://www.purplle.com";
    }

    public static boolean headless() { no usages
        return false;
    }
}
```

11. DriverFactory.java

Purpose:

- Creates WebDriver instance
- Configures Chrome browser

Key Features:

- ChromeOptions setup
- Headless execution support
- Window maximization

```

package com.example.Selenium.support;

import com.example.Selenium.config.Config;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.chrome.ChromeOptions;

public final class DriverFactory {  no usages

    private DriverFactory(){}  no usages

    public static WebDriver createChromeDriver() {  no usages

        ChromeOptions options = new ChromeOptions();

        if (Config.headless()) {
            options.addArguments("--headless=new");
        }

        options.addArguments("--window-size=1440,900");
        options.addArguments("--start-maximized");

        return new ChromeDriver(options);
    }
}

```

12. BaseTest.java

Purpose:

- Common test setup
- Browser initialization
- Browser cleanup

Key Features:

- @BeforeEach setup
- @AfterEach teardown

```

@BeforeEach
void setUp() {
    driver = DriverFactory.createChromeDriver();
}

@AfterEach
void tearDown() {
    if (driver != null) {
        driver.quit();
    }
}

```

13. HomePage.java

Purpose:

- Handles homepage interactions

Implemented Features:

- Open website
- Locate search bar
- Search products dynamically
- Press Enter using Actions class

Locators Used:

Search Box:

.search-input input

```

private final By searchBox = 1 usage
    By.cssSelector(".search-input input, input[placeholder*='Search']");

public void open(String url) { 1 usage
    driver.get(url);
}

```

14. SearchResultsPage.java

Purpose:

- Handles search result interactions

Implemented Features:

- Capture all products
- Print product list
- Click first product dynamically

Dynamic Locator Used:

a[id^='listing_item_']

This locator dynamically identifies all products.

15. ProductPage.java

Purpose:

- Handles product detail validations

Implemented Features:

- Capture product name
- Capture product price
- Capture product rating
- Add product to cart

Dynamic Locators Used:

Product Name:

h1

Product Price:

strong.our-price

Product Rating:

p.star-rating

Add To Cart:

button containing "Add to Cart"

16. CartPage.java

Purpose:

- Validate cart functionality

Implemented Features:

- Verify cart page loaded
- Validate cart count

Locator Used:

app-cart

17. Explicit Waits Used

The framework uses:

WebDriverWait

Benefits:

- Synchronization handling
- Dynamic element loading
- Stable automation execution

ExpectedConditions used:

- elementToBeClickable
 - visibilityOfElementLocated
 - presenceOfElementLocated
 - numberOfElementsToBeMoreThan
 - urlContains
-

18. Dynamic Locator Strategy

Dynamic locators were implemented to avoid hardcoded values.

Examples:

a[id^='listing_item_']

Benefits:

- Better maintainability
- Handles dynamic web elements
- Reduces locator failures

19. Assertions Used

JUnit Assertions:

- `assertTrue()`
- `assertFalse()`

Validations Performed:

- Product name not empty
- Product price not empty
- Cart page displayed
- Cart count greater than zero

```
String productName = productPage.getProductName();
String productPrice = productPage.getProductPrice();
String productRating = productPage.getProductRating();

System.out.println("Product Name : " + productName);
System.out.println("Product Price : " + productPrice);
System.out.println("Product Rating : " + productRating);

Assertions.assertFalse(productName.isEmpty());
Assertions.assertFalse(productPrice.isEmpty());

productPage.clickAddToCart();

Assertions.assertTrue(cartPage.isCartDisplayed());
Assertions.assertTrue( condition: cartPage.getCartCount() > 0);
```

20. Sample Console Output

Total Products Found : 8

1 -> Faces Canada Lipstick

2 -> Lakme Lipstick

3 -> Maybelline Lipstick

Product Name : Faces Canada

Comfy Matte Wow Liquid Lipstick - Cocoa Crush 07, 3.8 ml | One Swipe Application |
Highly Pigmented | Comfortable Wear | Glides Smoothly | Long Lasting | Transferproof

Product Price : ₹259

Product Rating : ★ 4.3

```
✓ 1 test passed 1 test total, 18 sec 211 ms
Nourishes & Keeps Lips Soft Pack of 3-414000
5 -> NY Bae Confessions Liquid Lipstick - Snuggle Secrets 15 (4.5 mL) | Matte Lipstick | Brown Shade | Moisturizing | Long Lasting
| Transfer Proof-373206
6 -> FACES CANADA Weightless Matte Lipstick - Royal Maroon 16 (Maroon), 4.5 gm | Highly Pigmented Lip Color | Smooth One Stroke
Glide | Moisturizes & Hydrates Lips | Vitamin E, Jojoba & Almond Oil Enriched-399931
7 -> NY Bae Super Matte Lipstick - Passionate Pamela 12 (4.2 gm) | Nude Pink | Loaded With Vitamin E | Rich Colour | Long Lasting |
Smudgeproof | Vegan-356176
8 -> Blue Heaven Intense Matte Lipstick 4gm Aged Wine S-308-415954
Product Name : Faces Canada
Comfy Matte Wow Liquid Lipstick - Cocoa Crush 07, 3.8 mL | One Swipe Application | Highly Pigmented | Comfortable Wear | Glides
Smoothly | Long Lasting | Transferproof
Product Price : ₹259
Product Rating : ★ 4.3
```

21. Test Execution

Command used:

mvn test

22. Expected Result

Expected Output:

BUILD SUCCESS

All validations should pass successfully.

```
Terminal Local x + -
6 -> FACES CANADA Weightless Matte Lipstick - Royal Maroon 16 (Maroon), 4.5 gm | Highly Pigmented Lip Color | Smooth One Stroke Glide | Moist
izes & Hydrates Lips | Vitamin E, Jojoba & Almond Oil Enriched-399931
7 -> NY Bae Super Matte Lipstick - Passionate Pamela 12 (4.2 gm) | Nude Pink | Loaded With Vitamin E | Rich Colour | Long Lasting | Smudgepro
| Vegan-356176
8 -> Blue Heaven Intense Matte Lipstick 4gm Aged Wine S-308-415954
Product Name : Faces Canada
Comfy Matte Wow Liquid Lipstick - Cocoa Crush 07, 3.8 mL | One Swipe Application | Highly Pigmented | Comfortable Wear | Glides Smoothly | Lo
Lasting | Transferproof
Product Price : ₹259
Product Rating : ★ 4.3
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 14.04 s -- in com.example.Selenium.tests.PurpleSearchTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 20.332 s
[INFO] Finished at: 2026-06-29T17:46:07+05:30
[INFO] -----
PS C:\Users\310970\Desktop\SDET TEST\today\KarthikeyanSeleniumTest29jun>
```

23. Advantages of Framework

- Reusable framework

- Scalable design
 - Dynamic element handling
 - Easy maintenance
 - Supports future enhancements
 - POM implementation
 - Explicit wait handling
-

24. Challenges Faced

During automation:

- Dynamic product loading
- Dynamic product locators
- Add-to-cart synchronization
- Search result handling

These were solved using:

- Explicit waits
 - Dynamic CSS selectors
 - Reusable method
-

25. Conclusion

The Selenium automation framework for Purple website was successfully implemented.

The automation successfully performed:

- Product search
- Product validation
- Product selection
- Cart validation

Github Link: [Branches · ust-sdet-training/SDET](#) Branchname: Karthikeyan_ApiSelenium_29jun

